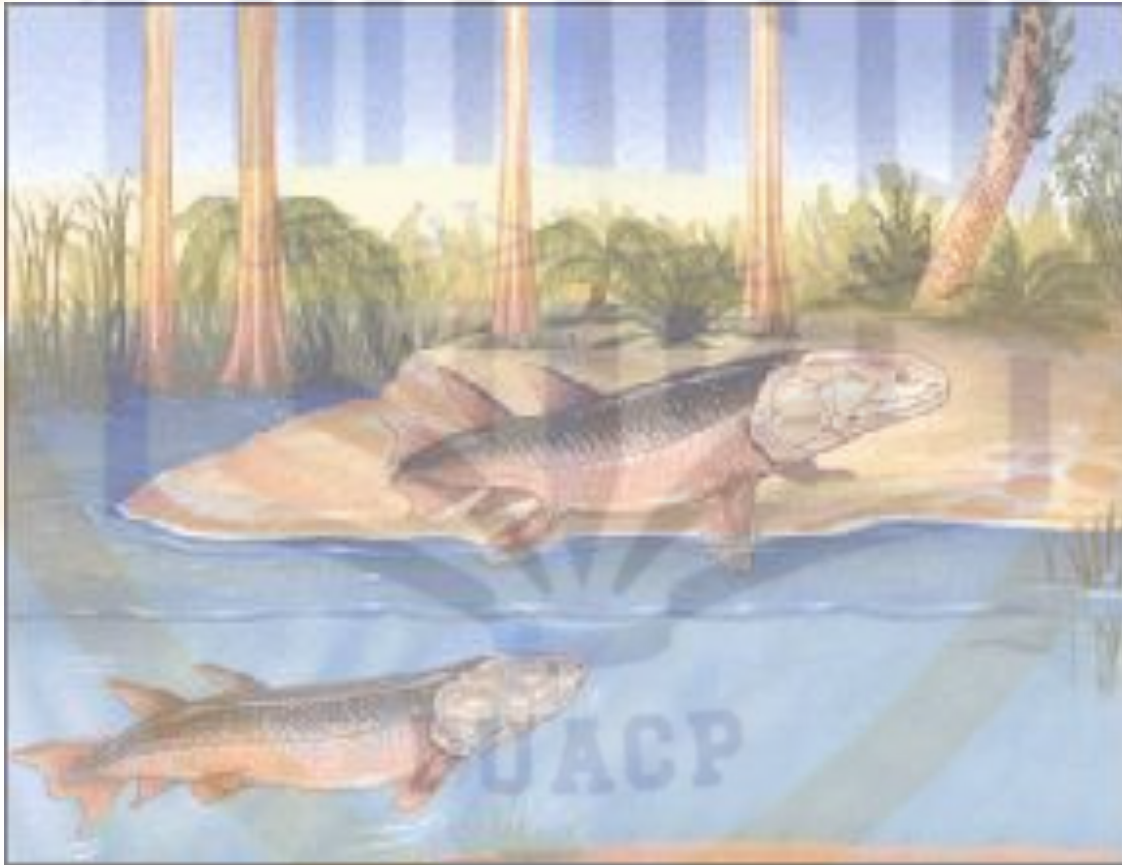


Lungs.....Lungfishes



- When fish are taken out of the water they suffocate, not because they can't breathe the oxygen available in the air
- Their gill arches collapse and there is not enough surface area for diffusion to take place
- Some fish species are designed to be exposed to the air for brief periods



Air Breathing Fish!?

- ♦ The walking catfish can go from pond to pond as long as their gills stay wet



walking catfish
Clarias batrachus
Photo by Chuck Cichra
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Lungs..... Lungfishes

Australian lungfish
(*Neoceratodus forsteri*)



African lungfish
(*Protopterus annectens*)



South American lungfish
(*Lepidosiren paradoxa*)



LUNGS

- Subclass Dipnoi
- living air-breathing fishes
- Characterized by the possession of either one or two lungs.

Australian lungfish
(*Neoceratodus forsteri*)



African lungfish
(*Protopterus annectens*)



South American lungfish
(*Lepidosiren paradoxa*)



Size, Number and location

- The **Australian lungfish**, may weigh up to 10 kg and grow to a length of 1.25 metres. **One lung... dorsal to the gut** but maintains a pneumatic duct that opens into the ventral wall of the gut.
- **African lungfishes**, is the largest, growing to a length of 2 metres. **Two lungs, bilobed but ventral to the gut.**
- The **South American species**, reaches a length of 1.25 metres (about 4 feet). **Two lungs, bilobed but ventral to the gut.**

Economic importance

- The economic importance of the lungfishes is slight.
- Only in certain parts of Africa, because of their abundance and size, are they of any value to humans as food.
- They are obtained from the mud of dried river bottoms.
- The South American lungfish, which is obtained in the same manner, is eaten locally.

Reproduction and life cycle

- The **African lungfishes** spawn in the last half of winter, the onset of the rainy season.
- *Protopterus* species build a nest in the form of a pit on the bottom of a watercourse.
- The egg is about 3.5 to 4 mm (about 0.14 inch) in diameter,
- The tiny larvae emerge a week after the eggs are laid.
- The larvae have long, bright red, tuftlike or **fanlike external gills**, which they use for breathing until the lungs are fully developed.
- The young at first remain in the nest under the protection of the male.

Reproduction and life cycle

- The **South American lungfishes** dig a nest in the bottom in the form of a vertical passage, which frequently turns horizontally at the bottom.
- The male remains in the nest and guards the brood. **During the spawning season, the pelvic fins of the male develop numerous tuft-shaped growths filled with small blood vessels (capillaries). These growths are believed to release oxygen from the blood, thereby oxygenating the water around the young.**
- The **Australian lungfish** lays gelatinous eggs among water plants; the larvae, which have **no external gills, breathe through internal gills.**

Behaviour and ecology

- Lungfishes are **voracious**, eating a variety of aquatic animals, including members of their own species.
- In captivity, African lungfishes eat earthworms, pieces of meat, tadpoles, small frogs, and small fish.
- The African lungfishes, has at the front of the upper jaw two rather rounded teeth with a hard transverse (from side to side) bridge. The lower jaw has a number of crushing teeth. The prey is sucked in, crushed, and thoroughly chewed; such a manner of eating is rare among fishes.

Adaptations for breathing

- There are a number of fishes that, in addition to or in place of gill breathing, have developed special organs through which they can breathe atmospheric air at the water surface. This occurs almost exclusively **in freshwater fishes**.
- In lungfishes these organs are, both in function and in structure, primitive lungs like those of amphibians. The name *lungfish* is thus well applied: these fishes have lungs that are **derived from the swim bladder** (an organ used for buoyancy in most bony fishes), which is connected to the alimentary tract.

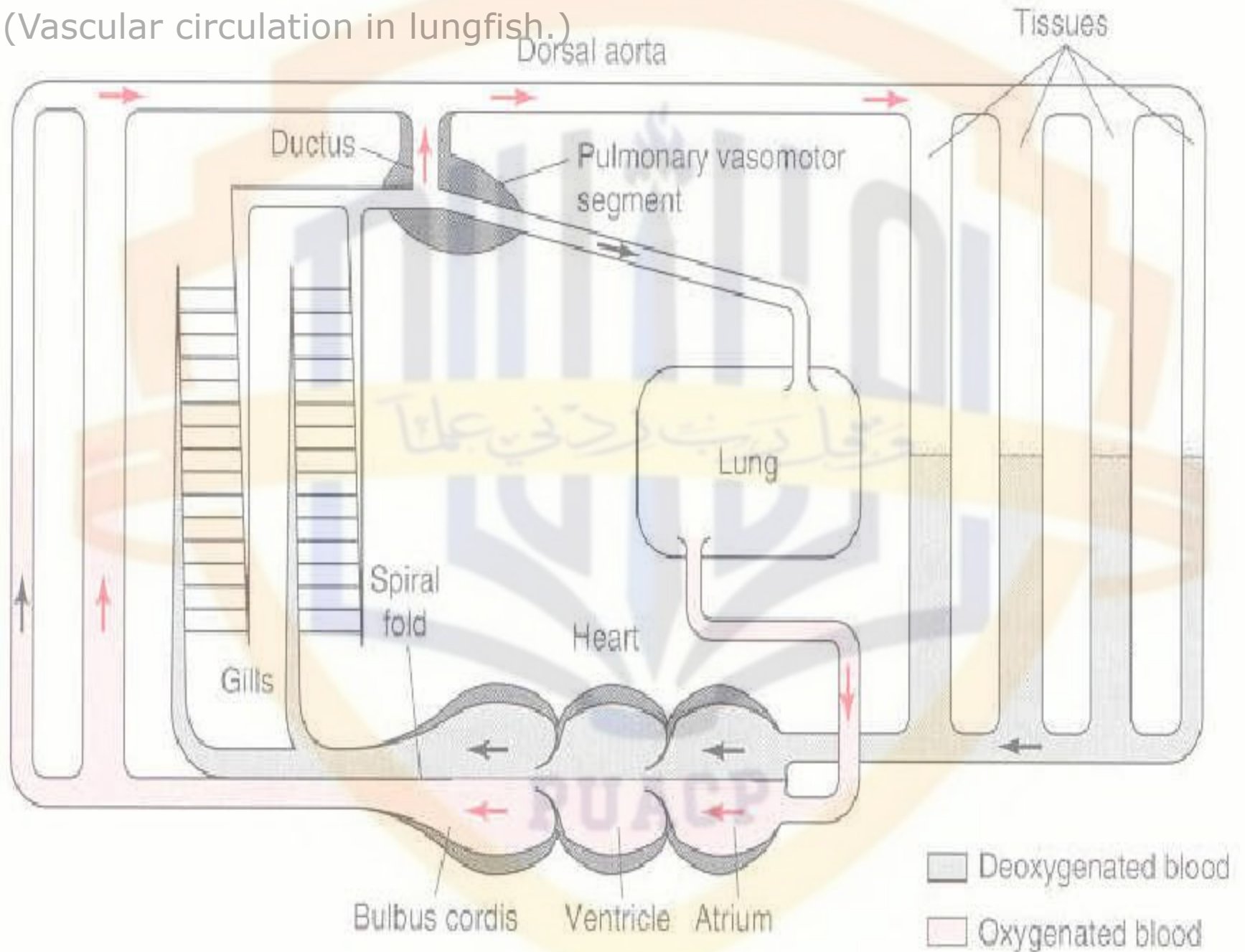
Adaptations for breathing

- The **internal septa ridges and pillars** divide the air spaces into **smaller compartments**
- Further, subdivisions of these compartments by septa produce alveoli like pockets covered with blood vessels.
- As in terrestrial higher vertebrates, gas exchange takes place in tiny alveoli like pockets (air vesicles).
- **Short pneumatic duct, lung paranchyma and smooth muscles** play role in **air distribution in lungs**.

Adaptations for breathing

- Structure of lungs in lungfishes not differ from higher vertebrates.
- The lungs originate as a **median ventral evagination of the embryonic foregut.**
- Also as in terrestrial vertebrates, there is a separate pulmonary circulation.
- Blood comes from **last efferent branchial artery** (fused third branchial artery in African lungfish and fourth in other lungfishes).
- Blood aerated in lungs returns to the heart directly through a **pulmonary vein into the left side of atrium.**

(Vascular circulation in lungfish.)



Adaptations for breathing

- In order to breathe, the fish swims upward and positions its head so that the tip of the snout barely touches the water surface. The mouth is then opened wide, and the fish sucks in air from just above the water—a process often accompanied by a characteristic sound.

Adaptations for breathing

- The **Australian lungfish** reportedly breathes air through the **nasal openings**, the mouth remain closed.
- In contrast to the more advanced bony fishes, lungfishes have a particular opening (choana) that connects the nasal cavity with the mouth.
- In the Australian lungfish, gill breathing predominates at least some of the time—namely, in times of normal water level when the water is well oxygenated. At such times the fish rises less often to the surface to breathe atmospheric air.

Adaptations for breathing

- When the water level goes down, which usually occurs in **August or September**, the fish is often found in isolated waterholes in which the oxygen content is greatly reduced.
- Other fishes in such pools often die from lack of oxygen, but the lungfish survives, having changed over to the breathing of atmospheric air.
- During such a dry period the Australian lungfish surfaces about every 40 to 50 minutes for air. African lungfishes surface for air about every 30 minutes or, in some cases, at longer intervals.

Adaptations for breathing

- African lungfishes estivate in a cocoon of hardened slime
- Animal rests coiled in cocoon with mouth upwards, its **lips connected by a funnel** to an air passage from the surface.
- Lungfish stays common for 3-4 months of dry season.
- During this period, metabolism reduced and oxygen consumption falls
- Due to rain, the funnel fills with water....air can no longer reach the lungs and animal awakens.
- Cocoon is ruptured and fish use its gills for the rest of the year.

Adaptations for breathing

- Presumably the South American lungfish estivates similarly.
- The Australian fish species survives only when in water. It neither burrows nor forms a cocoon.